

Calculus 2 - Homework 1 (Problems)

1.

Evaluate $\int 1/(x^2 - 4) dx$.

2.

Show that the Ratio Test is inconclusive for $\sum_{n=1}^{\infty} (n + 1)/(n^3 + 2)$, then determine convergence by another method.

3.

Evaluate $\int x/\sqrt{9 - x^2} dx$.

4.

Use a comparison test to determine whether $\sum_{n=1}^{\infty} n/(n^3 + 1)$ converges or diverges.

5.

Evaluate $\int \cos(x) e^{\sin x} dx$.

6.

Evaluate $\int \sqrt{x^2 + 1} dx$.

7.

Evaluate $\int 1/\sqrt{x^2 - 9} dx$ (for $x > 3$).

8.

Evaluate $\int x^2 e^x dx$.

9.

Evaluate $\int 1/\sqrt{9 - x^2} dx$.

10.

Use a comparison test to determine whether $\sum_{n=1}^{\infty} 1/(2n + 1)$ converges or diverges.